# Analysis I

## Uebungsblatt 8

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# 1

*Solution.* 1. Define the sequence $(x_n)_{n\in\mathbb{N}}$ by $x_n = \frac{1}{n}$.

- $\sum_{n=1}^{\infty} |x_n| = \sum_{n=1}^{\infty} \frac{1}{n} = \infty$ (Harmonic series).
- $\sum_{n=1}^{\infty} |x_n|^2 = \sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6} < \infty$.
- $\sup_{n\in\mathbb{N}} |x_n| = \sup_{n\in\mathbb{N}} \frac{1}{n} = 1 < \infty$.

Therefore: $(x_n) \notin \ell_1$ and $(x_n) \in \ell_2 \cap \ell_\infty$.

2. Define the sequence $(y_n)_{n\in\mathbb{N}}$ by $y_n = 1$.

- $\sum_{n=1}^{\infty} |y_n| = \sum_{n=1}^{\infty} 1 = \infty$.
- $\sum_{n=1}^{\infty} |y_n|^2 = \sum_{n=1}^{\infty} 1^2 = \infty$.
- $\sup_{n\in\mathbb{N}} |y_n| = \sup_{n\in\mathbb{N}} 1 = 1 < \infty$.

Therefore: $(y_n) \notin \ell_1 \cup \ell_2$ and $(y_n) \in \ell_\infty$. 8/8 □

# 2

## (a)

*Solution.* Since $\mathbb{R}^k$ is a finite-dimensional vector space, all norms are equivalent. Convergence in $(\mathbb{R}^k, \|\cdot\|_\infty)$ is therefore equivalent to convergence in $(\mathbb{R}^k, \|\cdot\|_1)$.

We check for convergence by examining component-wise convergence. Let $j \in \{0, \ldots, k-1\}$ be a fixed index. The $j$-th component of $a_n$ is $A_{n,j}$.

For $n \geq j$, we have $A_{n,j} = \frac{1}{n+j+1}$. We calculate the limit as $n \to \infty$:

$$\lim_{n\to\infty} A_{n,j} = \lim_{n\to\infty} \frac{1}{n+j+1} = 0$$

Since every component $j \in \{0, \ldots, k-1\}$ converges to 0, the vector sequence $(a_n)$ converges to the zero vector $L = (0, \ldots, 0) \in \mathbb{R}^k$.

(i) Yes, $(a_n)$ converges to the zero vector.

(ii) Yes, $(a_n)$ converges to the zero vector.

□

### (b)

#### (i)

*Solution.* We have

$$\|b_n\|_1 = \sum_{m=0}^{\infty} |A_{n,m}| = \sum_{m=0}^{n} \frac{1}{n+m+1},$$

which is a finite sum of $n+1$ positive terms, hence finite. Thus $b_n \in \ell_1$ for all $n$. □

#### (ii)

this you should show explicitly (for $\infty$ dim vector spaces)

*Solution.* For each fixed $m$, $\lim_{n\to\infty} A_{n,m} = \lim_{n\to\infty} \frac{1}{n+m+1} = 0$, so the only possible limit in $\ell_1$ is the zero sequence. But

$$\|b_n\|_1 = \sum_{m=0}^{n} \frac{1}{n+m+1} = \frac{1}{n+1} + \cdots + \frac{1}{2n+1} \geq (n+1)\cdot\frac{1}{2n+1} = \frac{n+1}{2n+1} \xrightarrow[n\to\infty]{} \frac{1}{2} \neq 0.$$

Hence $(b_n)$ does not converge in $\|\cdot\|_1$. □

[✓]

#### (iii)

*Solution.*

$$\|b_n\|_2^2 = \sum_{m=0}^{\infty} |A_{n,m}|^2 = \sum_{m=0}^{n} \left(\frac{1}{n+m+1}\right)^2 \leq (n+1)\left(\frac{1}{n+1}\right)^2 = \frac{1}{n+1} \xrightarrow[n\to\infty]{} 0.$$

Thus $\|b_n\|_2 \to 0$ and $(b_n)$ converges to 0 in $\|\cdot\|_2$. □

#### (iv)

*Solution.*

$$\|b_n\|_\infty = \sup_{m\geq 0} |A_{n,m}| = \max\left\{\frac{1}{n+1}, \frac{1}{n+2}, \ldots, \frac{1}{2n+1}\right\} = \frac{1}{n+1} \xrightarrow[n\to\infty]{} 0.$$

Thus $(b_n)$ converges to 0 in $\|\cdot\|_\infty$. □

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## 3

### (a) Boundedness of each $a_n$

*Solution.* For $m \geq 1$: $0 \leq A_{n,m} = \frac{m^n}{1+m^n} < 1$. For $m = 0$: $A_{n,0} = 0$. Hence $|a_n|_\infty = \sup_{m\in\mathbb{N}} |A_{n,m}| \leq 1 < \infty$, so $a_n \in \ell_\infty$ for all $n$. □

## (b) Cauchy property in $(\ell_\infty, \|\cdot\|_\infty)$

*Solution.* Let $n > k$. Then for $m \geq 2$

$$|A_{n,m} - A_{k,m}| = \left|\frac{1}{1+m^k} - \frac{1}{1+m^n}\right| \leq \frac{1}{1+m^k} \leq \frac{1}{1+2^k}.$$

For $m = 0, 1$ the difference is 0. Therefore

$$\|a_n - a_k\|_\infty = \sup_{m\in\mathbb{N}} |A_{n,m} - A_{k,m}| \leq \frac{1}{1+2^k}.$$

Given $\varepsilon > 0$, choose $N$ with $1/(1+2^N) < \varepsilon$. Then for all $n, k \geq N$, $\|a_n - a_k\|_\infty < \varepsilon$. Thus $(a_n)$ is Cauchy in $\ell_\infty$. □

## (c) Componentwise limit and membership in $\ell_\infty$

*Solution.* For $m = 0$: $L_0 = \lim_{n\to\infty} A_{n,0} = 0$. For $m = 1$: $L_1 = \lim_{n\to\infty} \frac{1}{1+1} = 1/2$. For $m \geq 2$: $L_m = \lim_{n\to\infty} \frac{m^n}{1+m^n} = 1$. Hence

$$L = (0, \tfrac{1}{2}, 1, 1, 1, \ldots), \qquad \|L\|_\infty = \sup_m |L_m| = 1 < \infty,$$

so $L \in \ell_\infty$. □

## (d) Convergence in the sup norm

*Solution.* For $m = 0, 1$ we have $A_{n,m} = L_m$. For $m \geq 2$,

$$|A_{n,m} - L_m| = \left|\frac{m^n}{1+m^n} - 1\right| = \frac{1}{1+m^n} \leq \frac{1}{1+2^n}.$$

Thus

$$\|a_n - L\|_\infty = \sup_{m\in\mathbb{N}} |A_{n,m} - L_m| = \frac{1}{1+2^n} \xrightarrow[n\to\infty]{} 0,$$

so $a_n \to L$ in $(\ell_\infty, \|\cdot\|_\infty)$. □

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## 4

*Solution.* Let $(x^{(k)})_{k\in\mathbb{N}}$ be a Cauchy sequence in $(\ell_\infty, \|\cdot\|_\infty)$.

Pointwise limit: For each fixed index $i \in \mathbb{N}$,

$$|x_i^{(k)} - x_i^{(l)}| \leq \|x^{(k)} - x^{(l)}\|_\infty,$$

so $(x_i^{(k)})_k$ is Cauchy in $\mathbb{K}$ and hence convergent. Define

$$x_i^* := \lim_{k\to\infty} x_i^{(k)}, \qquad x^* := (x_i^*)_{i\in\mathbb{N}}.$$

Uniform bound for $x^*$: Since $(x^{(k)})$ is Cauchy, choose $k_0$ with $\|x^{(k)} - x^{(l)}\|_\infty < 1$ for all $k, l \geq k_0$. Then for $k \geq k_0$, $\|x^{(k)}\|_\infty \leq \|x^{(k_0)}\|_\infty + 1$. Set

$$M := \max\{\|x^{(1)}\|_\infty, \ldots, \|x^{(k_0-1)}\|_\infty,\ \|x^{(k_0)}\|_\infty + 1\}.$$

For every $i$, $|x_i^*| = \lim_{k\to\infty} |x_i^{(k)}| \leq M$. Hence $x^* \in \ell_\infty$ and $\|x^*\|_\infty \leq M$.

Convergence in the sup norm: Let $\varepsilon > 0$. Pick $k_1$ such that $\|x^{(k)} - x^{(l)}\|_\infty < \varepsilon$ for all $k, l \geq k_1$. Fix $k \geq k_1$ and let $l \to \infty$. For every $i$,

$$|x_i^{(k)} - x_i^*| = \lim_{l\to\infty} |x_i^{(k)} - x_i^{(l)}| \leq \varepsilon.$$

Taking the supremum over $i$ gives $\|x^{(k)} - x^*\|_\infty \leq \varepsilon$ for all $k \geq k_1$. Thus $x^{(k)} \to x^*$ in $\ell_\infty$.

Therefore every Cauchy sequence in $(\ell_\infty, \|\cdot\|_\infty)$ converges; the space is complete. $\square$

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